

SECURITY CLEARANCE: MANUAL-ONLY EXPORT

PART 1: GLOBAL DIMENSIONS & MATHEMATICAL TOPOLOGY

System Identification: AELV-1M-AF / S-M-Platform (Corrected Geometry Specs)

Author Designation: User

Target Interpreter: Muse (3D-Generation & Object Synthesis)

1.1. Modeling Directives for Object Generation

- Strict Planar Geometry:** Do not use NURBS, splines, fillets, or chamfers. All surfaces must be planar polygons. All intersections must result in hard, sharp edges (Material-on-Material).
- Extrusion Logic:** The hulls are modeled as 2D vertical extrusions to maintain strict hydrodynamic Venturi ratios across all draught levels.
- Units & Scaling:** All Cartesian coordinates (X, Y, Z) are defined in millimeters (mm).
- Origin Point** $(0, 0, 0)$: Located at the bottom-center of the stern (rear).
 - $+X$ = Starboard (Right)
 - $-X$ = Port (Left)
 - $+Y$ = Bow (Forward)
 - $+Z$ = Zenith (Up)

1.2. Global Bounding Box & Solar Deck

To meet the 1.8 m^2 requirement for the 396W solar array, the overall dimensions are rigidly locked:

- Total Length** (L_{max}): 1500 mm
- Total Width** (W_{max}): 1200 mm
- Total Height** (H_{max}): 400 mm
- Nominal Draught** (D_{nom}): 200 mm (Waterline is at $Z = 200$)

The Solar Deck is a single continuous planar sheet capping the structure, directly resting on the sharp upper edges of the hulls:

$$\text{Deck}_{\text{Area}} = L_{max} \cdot W_{max} = 1500 \text{ mm} \cdot 1200 \text{ mm} = 1.8 \cdot 10^6 \text{ mm}^2 = 1.8 \text{ m}^2$$

Deck Vertices (X, Y, Z) :

- $(-600, 0, 400)$
- $(600, 0, 400)$
- $(600, 1500, 400)$
- $(-600, 1500, 400)$

PART 2: HULL PONTOONS & VENTURI CHANNEL COORDINATES

2.1. Venturi Ratio Verification (Correction from Sketch)

The inner walls of the catamaran form the acceleration channel for the AELV-1M fibers. The sketch incorrectly showed a parallel 600 mm channel. To achieve the derived velocity amplification ratio of 1.6, the channel width must decrease linearly from the bow to the throat.

- **Inlet Width** (W_{inlet} at $Y = 1500$): 640 mm
- **Throat Width** (W_{throat} at $Y = 750$): 400 mm

$$\frac{A_1}{A_2} = \frac{W_{inlet} \cdot D_{nom}}{W_{throat} \cdot D_{nom}} = \frac{640}{400} = 1.6$$

2.2. Starboard Hull (Right Pontoon)

This is a solid polygonal prism. All vertical edges are strictly parallel to the Z-axis. The base polygon at $Z = 0$ is identically extruded to $Z = 400$. Model this as a solid volume for boundary definition.

Starboard Base Polygon Vertices at $Z = 0$ (Extrude linearly to $Z = 400$):

- P_1 (Inner Stern): (200, 0, 0)
- P_2 (Inner Throat): (200, 750, 0)
- P_3 (Inner Bow / Sharp Knife Edge): (320, 1500, 0)
- P_4 (Outer Bow Shoulder): (600, 1200, 0)
- P_5 (Outer Stern): (600, 0, 0)

Geometric Note for Muse: The outer wall runs straight along $X = 600$ from $Y = 0$ to $Y = 1200$. From $Y = 1200$ to $Y = 1500$, it cuts inward to meet the inner wall at $X = 320$, forming a sharp hydrodynamic wedge with an entry angle of:

$$\theta = \arctan\left(\frac{600 - 320}{1500 - 1200}\right) = \arctan\left(\frac{280}{300}\right) \approx 43.02^\circ$$

2.3. Port Hull (Left Pontoon)

Exact mirror of the Starboard hull across the Y-Z plane ($X = 0$).

Port Base Polygon Vertices at $Z = 0$ (Extrude linearly to $Z = 400$):

- P_6 (Inner Stern): (-200, 0, 0)
- P_7 (Inner Throat): (-200, 750, 0)
- P_8 (Inner Bow / Sharp Knife Edge): (-320, 1500, 0)
- P_9 (Outer Bow Shoulder): (-600, 1200, 0)
- P_{10} (Outer Stern): (-600, 0, 0)

PART 3: SUBSYSTEM INTEGRATION VOLUMES

To assemble the final design, Muse must place the operational hardware within the central void (the Venturi channel). These volumes represent the absolute physical boundaries for the hardware modules to ensure the 1.6 ratio and flow mechanics remain completely undisturbed. All component housings must utilize hard, orthogonal angles.

3.1. Front Module: AELV-1M Fiber Dispenser

Located at the top of the inlet, dropping fibers into the incoming accelerated wave crest.

- **X-Axis:** $[-150, 150]$ (Centered, 300 mm width)
- **Y-Axis:** $[1300, 1400]$ (100 mm behind the bow tip)
- **Z-Axis:** $[250, 400]$ (Hanging below the solar deck, completely above the 200 mm waterline)

3.2. Center Module: Electric Jet Drive

Positioned perfectly in the throat to induce the required 5.42 m/s exhaust velocity. Sits fully submerged.

- **X-Axis:** $[-75, 75]$ (Housing width 150 mm)
- **Y-Axis:** $[600, 750]$ (Mounted exactly at the narrowest section of the Venturi channel)
- **Z-Axis:** $[50, 200]$ (Fully submerged below the 200 mm waterline)

3.3. Core Module: AELV-1M Retrieval Cassette

The main storage block for the harvested fibers. Mounted centrally behind the jet, utilizing the jet's waste heat for thermal drying.

- **X-Axis:** $[-190, 190]$ (380 mm width, spanning almost the entire 400 mm parallel aft channel)
- **Y-Axis:** $[150, 500]$ (350 mm length)
- **Z-Axis:** $[150, 400]$ (Suspended mostly above water, base dipping slightly into the surface)

3.4. Stern Module: Electromagnetic Rail (PWM-Controlled)

The 0.8 T pulsed electromagnet. Must be submerged to pull the MK-35 cores from a depth of 200 mm.

- **X-Axis:** $[-190, 190]$ (380 mm width)
- **Y-Axis:** $[0, 100]$ (Flush with the absolute rear stern)
- **Z-Axis:** $[0, 150]$ (Submerged, acting as the hydrodynamic scraper plate)
- **Scraper Edge Vector:** A physical hard edge constructed at $Z = 150, Y = 0$ pointing towards $+Y$ to detach fibers from the magnetic field as they are mechanically pulled upward into the cassette.

Design Integrity Check:

No hardware component in the central channel exceeds $X = \pm 190$ in the aft section. This rigorously preserves a 10 mm clearance on each side against the inner PE-HD walls ($X = \pm 200$) to prevent mechanical binding and friction, fulfilling the cost-effective assembly directives.